Randomized and low-rank approximation

Convexity of the expected error of volume-sampled column subsets

Difficulty: challenging **Importance:** interesting to the community

Status: Solved

Rating rationale: Challenging because the required second-difference inequality goes beyond standard log-concavity; community impact is understanding average column-selection and Nyström error.

Resolution — 2026-09-11

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The sequence $(j+1)e_{j+1}/e_j$ is decreasing and discretely convex for every positive spectrum, including both endpoints. The exact second-difference certificate proves the full canonical conjecture.

The complete target is resolved. Its former difficulty rating is historical; the original mathematical statement and source evidence are retained below.

Primary reference: complete authored PDF, standalone TeX, **Theorem 1.1**. Independent proof review · Authorship and submission record. Verification is independent agent review, not external human peer review or formal certification.

For $n \geq 3$ and positive real numbers $\lambda_1, \ldots, \lambda_n$, let

$$e_j(\lambda) = \sum_{\substack{S \subseteq \{1,\ldots,n\} \\ |S|=j}} \prod_{i \in S} \lambda_i, \qquad e_0 = 1, \quad e_{n+1} = 0.$$

Define $f(j) = (j+1)e_{j+1}(\lambda)/e_j(\lambda)$ for $1 \leq j \leq n$. Is f discretely convex, that is,

$$f(j-1) - 2f(j) + f(j+1) \geq 0 \qquad (2 \leq j \leq n-1),$$

for every such n and positive tuple λ?

For a matrix A with positive eigenvalues λ_i of $A^T A$, sample a set S of j columns with probability proportional to $\det(A_S^T A_S)$. Its expected squared Frobenius projection error is

$$\mathbb{E}\|(I - P_{A_S})A\|_F^2 = f(j),$$

where P_{A_S} is the orthogonal projector onto the selected column span. Thus the conjecture asserts a diminishing marginal improvement in the average reconstruction error of these fixed-size determinantal samples. It is distinct from optimizing a worst-case approximation factor over all matrices with a prescribed spectrum.

References

1. M. Dereziński, R. Khanna, and M. W. Mahoney, *Improved guarantees and a multiple-descent curve for Column Subset Selection and the Nyström method*, NeurIPS 2020, arXiv:2002.09073. Section 5, Conjecture 1 is exactly the discrete-convexity statement. Paper.
2. A. Deshpande, L. Rademacher, S. S. Vempala, and G. Wang, *Matrix Approximation and Projective Clustering via Volume Sampling*, Theory of Computing 2 (2006), Article 12, pp. 225–247. See the volume-sampling matrix approximation results. Paper.

Status check — 2026-09-08

Searched "Derezinski Khanna Mahoney Conjecture 1 convexity", "elementary symmetric ratio $k+1$ convexity DPP", and the exact 2020 paper title with 2025/2026. Checked the current arXiv record. No proof or counterexample was found. Standard Newton inequalities imply other ratio inequalities but are not cited as a resolution of this second-difference inequality.

Audit — 2026-09-10

Rechecked Dereziński–Khanna–Mahoney, §5, Conjecture 1; the current record remains v3 of December 2020. Volume-sampling convexity and elementary-symmetric-ratio searches found no resolution. This remains historical explicit conjecture evidence, supplemented by a bounded follow-up search.